

NLU course project: Intent Classification and Slot Filling

Elena Deidda (mat. 267887)

University of Trento

elena.deidda@studenti.unitn.it

1. Introduction

I tackle joint intent classification and slot filling on ATIS, evaluated with Intent Accuracy and chunk-level Slot F1 (CoNLL). In Part A I train a GPT-2-style decoder-only Transformer *from scratch*, reaching test Slot F1 0.888 and Intent Accuracy 0.934. In Part B I fine-tune pre-trained BERT [1] and GPT-2 [2], following the joint intent/slot setup of [3]; the best model (BERT-base) reaches test Slot F1 0.955 and Intent Accuracy 0.975, confirming the benefit of a pre-trained, bidirectional encoder for slot tagging.

2. Implementation details

Slot loss (cross-entropy, padding ignored) and intent loss (cross-entropy) are summed; all models are run with multiple seeds and I report mean $\pm$ std on the test set. Hyper-parameters are always selected on the *dev* Slot F1.

Part A. Since the model is autoregressive, a CLS token is appended at the *end* of each utterance (not at the start): being the last position, it is the only one that has attended to the whole sentence, so its final hidden state is used for intent classification. The search is greedy: learning rate first, then the architecture (`d_model`, heads, layers, FFN size, each tested against the current incumbent), then dropout. Only `n_heads` (2 $\rightarrow$ 4) improved over the initial small model; `d_model`, `num_layers`, `ff_dim` and dropout were all tested but none of their candidates beat the incumbent dev F1, so the final model keeps the original (small) values for those hyper-parameters. I interpret this as ATIS being small enough that the smallest architecture already has sufficient capacity, with extra parameters or regularization providing no further benefit.

Part B. BERT uses the hidden state at [CLS] (first token) for intent classification, while GPT-2, being causal, uses the hidden state at the last *real* token (via the per-sequence length, not the last padded position). Sub-word labels are aligned so only the first sub-token of each word carries the real slot label; the rest (and the special tokens) are ignored in the loss. For each family I search learning rate then dropout on the base model, then evaluate the larger variant (BERT-large, GPT-2-medium) with the same configuration.

3. Results

Table 1 reports Part A: the lr search already reaches most of the achievable performance, and only `n_heads` contributes a further, modest gain. Table 2 reports Part B: BERT clearly outperforms GPT-2 on slot filling (0.955 vs 0.915 test F1), consistent with BERT’s bidirectional context being more suited to per-token tagging than GPT-2’s left-to-right attention. Scaling up to the large/medium variant does not improve test metrics for either family despite $3\times$ more parameters, suggesting ATIS is already saturated by the base-sized pre-trained models.

Table 1: *Part A — best model per step (231K param. throughput). Final: $d=64$, 4 heads, 2 layers, FFN 256, dropout 0.*

Step	Adopted	Dev F1	Test Slot F1	Test Int. Acc
0 – lr	0.01	0.929	0.876 $\pm$.007	0.921 $\pm$.017
1 – architecture	heads=4	0.941	0.888$\pm$.006	0.934$\pm$.006
2 – dropout	none (0)	0.941	0.888 $\pm$.006	0.934 $\pm$.006

Table 2: *Part B — fine-tuning (best lr/dropout per family).*

Model	Param	Dev F1	Test Slot F1	Test Int. Acc
BERT-base	109.6M	0.984	0.955$\pm$.002	0.975$\pm$.001
BERT-large	335.3M	0.983	0.955 $\pm$.001	0.975 $\pm$.002
GPT-2-base	124.6M	0.963	0.915 $\pm$.004	0.971 $\pm$.002
GPT-2-medium	355.0M	0.964	0.913 $\pm$.002	0.969 $\pm$.003

4. References

- [1] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proceedings of NAACL-HLT*, 2019.
- [2] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever, “Language models are unsupervised multitask learners,” OpenAI, Tech. Rep., 2019.
- [3] Q. Chen, Z. Zhuo, and W. Wang, “BERT for joint intent classification and slot filling,” in *arXiv preprint arXiv:1902.10909*, 2019.